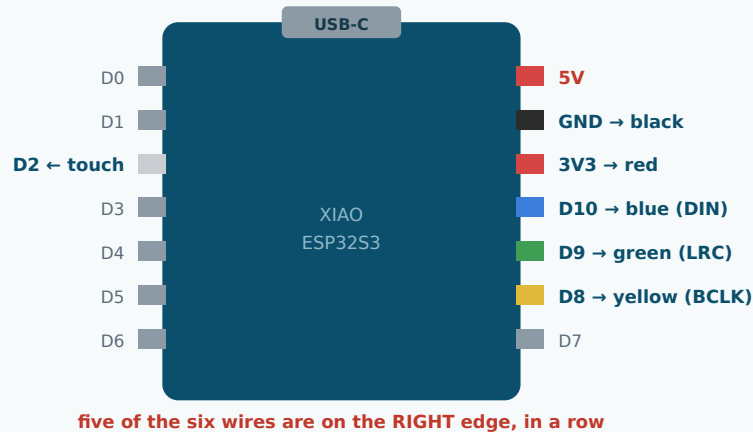


Wiring Guide

Eight joints. Read page 1 before you touch the iron.

READ THIS FIRST — the pad labels are not the GPIO numbers. On the XIAO ESP32S3 the silkscreen says D0 ... D10, but the code refers to GPIO numbers. **They do not match.** The code says GPIO 7, which is the pad printed D8 — not D7. Wire by the **PAD** column below, never by the GPIO number.

XIAO ESP32S3 — seen from above, USB at the top



Everything except the touch wire lands on one edge. That's convenient — solder them as a group.

SOLDER TO PAD	code calls it	Wire	Goes to
3V3	—	Red	Amp VIN
GND	—	Black	Amp GND
D8	GPIO 7	Yellow	Amp BCLK / SCK
D9	GPIO 8	Green	Amp LRC / WS
D10	GPIO 9	Blue	Amp DIN
D2	GPIO 3	White	Metal disc in the front lid

Verify this table against Seeed's own pinout diagram before soldering. It's the one thing here that's cheap to check and expensive to get wrong. If your board's diagram disagrees, believe the diagram and tell me — I'll correct pins.h rather than have you rewire.

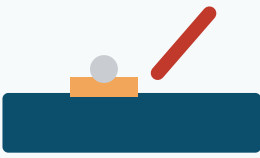
1

How to actually solder this

If you've never done it, start here

Soldering is **gluing with metal that conducts**. You heat the pad, feed in solder, it melts and sticks to both the pad and the wire. That's all.

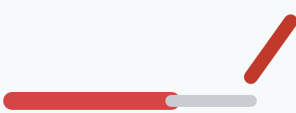
A Tin the pad



iron on the pad, 2 seconds
feed a little solder in

leaves a shiny blob

B Tin the wire



strip 3mm, twist the strands
heat, feed solder, it wicks in

tip goes stiff and silver

C Join them



hold wire on pad, touch iron
1 second — both blobs merge

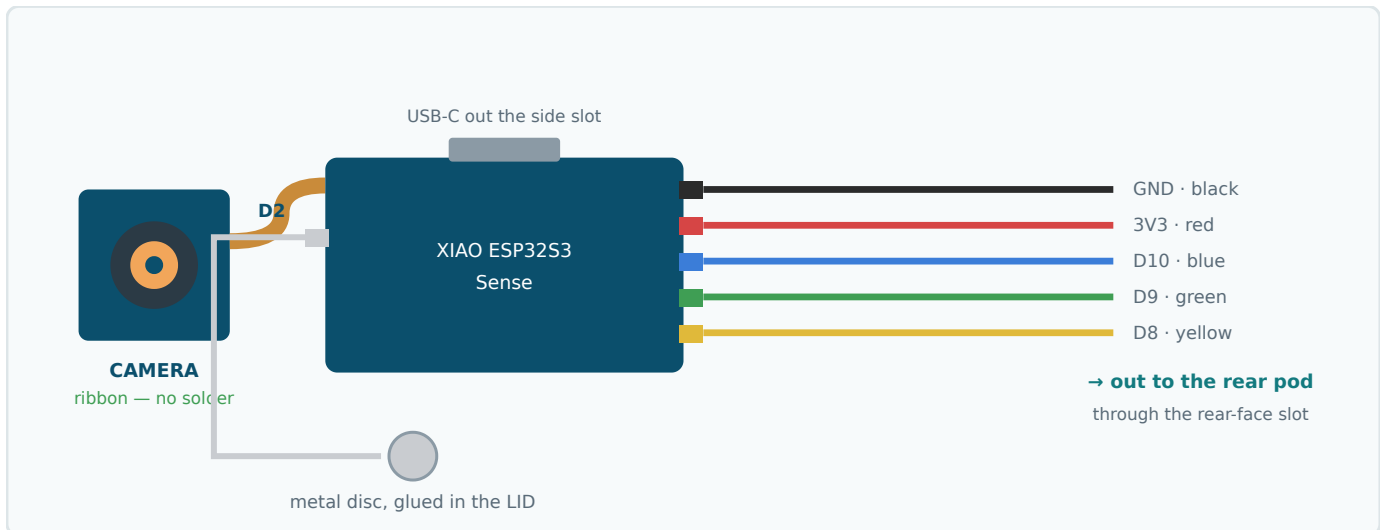
remove iron, don't move for 3s

Tin both sides separately, then just melt them together. Far easier than doing it in one go.

What you see	What it means
Shiny, smooth, slightly concave	Good joint. Move on.
Dull, grey, lumpy	Moved while cooling, or not hot enough. Re-heat until it flows, then hold still.
Ball sitting on top, not spread	The pad wasn't hot — you melted solder onto a cold pad. Heat the <i>pad</i> , not the solder.
Blob bridging two pads	Short circuit. Remove with braid or by dragging a clean hot tip across it.

Heat the pad, not the solder. That single sentence is 90% of learning to solder. Touch the iron to the pad, count two, *then* bring the solder in so the pad melts it. Melting solder on the tip and dabbing it on makes a joint that looks fine and falls off later.

Don't have an iron yet? The camera plugs in by ribbon — you can flash the board, run the server and read real labels today with zero soldering. See [Vocalens_NoSolder_Day.pdf](#) .



Five wires leave the pod as a bundle. The white touch wire stays inside.

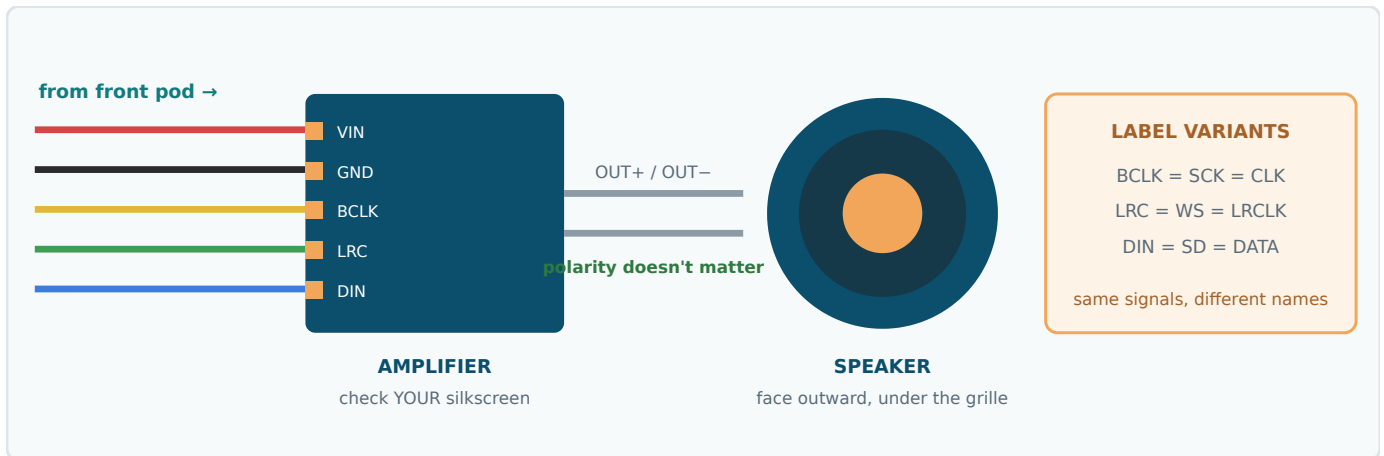
- 1 **Cut the five long wires first.** Measure your glasses from hinge to behind the ear, add 2cm. Too short can't be fixed.
- 2 Tin all six pads: 3V3 , GND , D8 , D9 , D10 , D2 .
- 3 Solder the five bundle wires to the right-hand edge. Work top to bottom so your iron never crosses a finished joint.
- 4 Solder the white wire to D2 on the left edge.
- 5 **Check for bridges.** Look along the pad row at a low angle — a bridge is obvious side-on and invisible from above.
- 6 Tug each wire gently. A joint that fails this on the bench will definitely fail on a face.

Keep the joints flat. Anything standing proud eats into the 10mm the pod has. If a blob is tall, re-heat it and let it settle rather than forcing the lid.

3

Rear pod — amp and speaker

Two more joints, then you're done



Five wires in, two wires out to the speaker. Nothing else in this pod.

Wire arriving	From XIAO pad	To amp pin
Red	3V3	VIN
Black	GND	GND
Yellow	D8	BCLK / SCK
Green	D9	LRC / WS
Blue	D10	DIN

If it's too quiet, try 5V instead of 3V3. Most I2S amp modules run on 3–5V and get noticeably louder at 5V. Check your module is rated for it first, and note the XIAO's 5V pad only carries 5V while USB is plugged in — on battery it's dead. Start on 3V3; it's the safe one.

Many amp modules have extra pins — GAIN, SD, SHUTDOWN. Leave them unconnected. They have sensible defaults, and SD in particular usually means "shutdown" — grounding it silences the amp, which is a confusing hour to lose.

#	Test	Passes when
1	Continuity, power OFF Multimeter on beep mode	Each pad beeps to its far end, and no pad beeps to its neighbour. Neighbours beeping = a bridge.
2	Power, nothing else	Plug in USB. Board enumerates, nothing gets hot. Anything warm — unplug immediately.
3	Camera alone	Server receives a frame at <code>/read_label</code> . No audio yet.
4	Audio, short wires	Reply plays through the speaker. Prove it before committing to a 10cm run.
5	Full-length wires, clipped on	Repeat 3 and 4 with everything mounted.
6	Touch pad	Tapping triggers. Tune <code>TOUCH_THRESHOLD</code> <i>while worn</i> , not on the bench.

Test 1 costs two minutes and catches the mistake that kills boards. A solder bridge between `3V3` and `GND` is a dead short. Find it with a meter before you find it with smoke.

TROUBLESHOOTING

Symptom	Cause
Camera never initialises	PSRAM not set to OPI. Check this first, every time.
No sound at all	Amp unpowered. Meter <code>VIN</code> – <code>GND</code> , expect ~3.3V.
Static or buzzing only	Yellow and green swapped. <code>D8</code> → <code>BCLK</code> , <code>D9</code> → <code>LRC</code> .
Silence, but wiring checks out	An <code>SD</code> /shutdown pin pulled low. Disconnect it.
Very quiet but correct	Speaker not under the grille, or try 5V.
Touch fires by itself	<code>TOUCH_THRESHOLD</code> needs tuning while worn.
Touch never fires	White wire not contacting the disc — scuff and re-solder.
Works on bench, fails when worn	A wire pulling at a pod exit. Add strain relief.
Board won't enumerate at all	Unplug everything. If it enumerates bare, you have a short.

Eight joints total. Six on the board, two on the speaker. If you can make one good joint you can make eight — and the tinning trick on page 2 makes each one about a second of actual iron contact.